Head Trauma

TB-05 | Trauma & Burns

Quick flow

01 Head Injury [EMT / AEMT / Paramedic]

Assess mechanism, loss of consciousness, anticoagulants, seizure, vomiting, and baseline.

02 Primary Survey [EMT / AEMT / Paramedic]

Protect airway and spine as indicated; control bleeding without compressing unstable skull injury.

03 GCS + Pupils [EMT / AEMT / Paramedic]

Document GCS components, pupils, focal deficits, and repeated neurologic examinations; use continuous SpO_2 and serial blood pressure monitoring.

04 Prevent Secondary Injury [EMT / AEMT / Paramedic]

Continuously monitor oxygenation and ventilation; aim for SpO_2 ≥94% and urgently correct SpO_2 <90% or inadequate ventilation. During initial and routine assisted ventilation, target $EtCO_2$ 35-45 mmHg; do not routinely hyperventilate. Maintain age-appropriate perfusion, glucose, and temperature. For suspected TBI, prevent hypotension: target adult SBP at least 110 mmHg, including age 70; do not use permissive hypotension. Reassess blood pressure frequently and after treatment; for children, obtain Medical Control for age-specific TBI resuscitation guidance rather than substituting an adult target.

05 Herniation Signs? [AEMT / Paramedic]

Falling GCS • unilateral dilated pupil • abnormal posturing • Cushing response. Position the head midline and elevate about 20-30° when this does not worsen hypotension or interfere with spinal precautions. Only for clear active or imminent herniation, use brief controlled rescue ventilation to $EtCO_2$ 30-35 mmHg with capnography while expediting transport; avoid overshooting below 30 mmHg; no routine hyperventilation.

06 Appropriate Destination [EMT / AEMT / Paramedic]

Minimize scene time, provide serial GCS and early notification, especially for neurologic deterioration or anticoagulant/antiplatelet use. Apply the trauma field-triage and destination plan: high-risk patients to the highest-level appropriate trauma center within regional constraints; favor pediatric-capable trauma care for children. Use a closer stabilization hospital only for an immediate critical need that cannot safely be managed during longer transport, under the destination plan/Medical Control. Notify early, document the reason, and activate the predefined transfer process. Do not delay ground movement solely while awaiting aircraft.

Overview

- Prevent secondary brain injury by avoiding hypoxia, hypotension, hyperventilation, and transport delay.
- Anticoagulant or antiplatelet use, age, mechanism, loss of consciousness, vomiting, seizure, and neurologic change increase risk.

Provider-Level Actions

Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT

- Maintain airway; continuously monitor oxygenation and ventilation, aim for SpO_2 ≥94%, and urgently correct SpO_2 <90% or inadequate ventilation. Use spinal motion restriction when indicated, control bleeding, obtain serial GCS/pupils, and transport promptly.
- Identify anticoagulants, antiplatelets, seizure, repeated vomiting, skull signs, focal deficit, abnormal pupils, and neurologic deterioration.

AEMT

- Perform all EMT care plus vascular access and age-appropriate treatment of hypotension without delaying transport; use AM-05 for adults. For suspected TBI, target adult SBP at least 110 mmHg, including age 70; do not use permissive hypotension. Reassess blood pressure frequently and after treatment; for children, obtain Medical Control for age-specific TBI resuscitation guidance rather than substituting an adult target.
- Use authorized airway adjuncts and $EtCO_2$ monitoring when available. For initial and routine assisted ventilation, target $EtCO_2$ 35-45 mmHg; avoid routine hyperventilation.

Paramedic

- Perform all prior care plus advanced airway management and controlled ventilation when the patient cannot protect the airway; for initial and routine assisted ventilation, including after advanced airway placement, target $EtCO_2$ 35-45 mmHg.
- Only for clear active or imminent herniation, use brief controlled rescue ventilation to $EtCO_2$ 30-35 mmHg with capnography while expediting transport; avoid overshooting below 30 mmHg; avoid routine hyperventilation. Treat active seizure under the seizure/medication pathway; do not provide prophylactic antiseizure medication in the field.

Indications

- Patient with suspected head trauma requiring prehospital assessment or treatment.

Warnings / precautions

- For pediatric TBI, obtain Medical Control guidance on age-specific perfusion targets. Do not substitute an adult target or a generic medical-shock threshold.
- A single normal examination does not exclude intracranial hemorrhage, especially in anticoagulated or older patients.
- Do not aggressively control hypertension in suspected traumatic brain injury unless another approved protocol requires it. Initial and routine assisted-ventilation target: $EtCO_2$ 35-45 mmHg. Brief rescue target: $EtCO_2$ 30-35 mmHg only for clear active or imminent herniation, with capnography and expedited transport; avoid overshooting below 30 mmHg and do not routinely hyperventilate.

Clinical pearls

- Hypotension in isolated head injury should prompt a search for extracranial hemorrhage.
- Trend the individual eye, verbal, and motor GCS components rather than documenting only a total score.

Related pathways / contacts

AM-05 Hypotension / Shock - Adult shock care with the higher suspected-TBI perfusion target.

Source and references

Current Tennessee EMS scope of practice, verified individual credentials, and Claiborne County standing orders control provider authorization.

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